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Ch-1: Project Definition :

Analysis of VNSGU Student Academic Performance using Python

Understanding student academic performance is essential for evaluating educational quality, identifying learning gaps, and improving institutional decision-making. In this project, we analyse the VNSGU student result dataset using Python to extract meaningful academic insights through Exploratory Data Analysis (EDA).

This project focuses on analysing structured examination result data to understand performance distribution, subject-wise trends, grading patterns, and overall academic outcomes.

The primary objective of this analysis is to transform raw examination result data into actionable insights using statistical techniques and data visualization tools.

Project Objectives :

We aim to answer the following analytical questions:

1. What is the overall performance distribution of students?
2. What is the average marks obtained in each subject?
3. Which subject has the highest failure rate?
4. How are total marks distributed among students?
5. What percentage of students passed and failed?
6. Identify top-performing students based on total marks.
7. Are there any outliers in subject-wise marks?
8. What is the correlation between subjects and total marks?

Ch-2: Dataset Details :

Dataset Name: VNSGU Master Student Result Dataset

Dataset Source:

<https://ums.vnsgu.net/Result/StudentResultDisplay.aspx?HtmlURL=7596,0000>

Dataset Type: Structured Excel Dataset (.xlsx)

Dataset Domain: Educational Data Analytics / Academic Performance Analysis

Dataset Description

The dataset contains academic performance records of students enrolled under VNSGU. It includes structured information related to student identification, subject-wise marks, total marks, and result status.

The dataset typically contains the following types of attributes:

- Seat Number (Unique Student Identifier)
- Student Name
- Collage Name
- Subject-wise Marks
- Total Marks
- SGPA
- Percentage
- Result Status (Pass/Fail)

Ch-3: Python tools and libraries used :

Python 3.10 :

Python is a programming language widely used by Data Scientists. Python has in-built mathematical libraries and functions, making it easier to calculate mathematical problems and to perform data analysis.

Pandas :

Pandas is a fundamental, open-source Python library for data manipulation and analysis, providing fast, flexible, and expressive data structures like DataFrames (2D tables) and Series (1D arrays) to handle tabular data efficiently, making it ideal for cleaning, transforming, exploring, and analyzing large datasets in data science, machine learning, and analytics. It simplifies tasks like reading/writing files (CSV, Excel), handling missing data, filtering, grouping, merging, and performing statistical analysis, integrating seamlessly with other tools like NumPy and Matplotlib.

NumPy :

NumPy (Numerical Python) is a foundational Python library for data science, primarily used for its efficient multi-dimensional array object (the ndarray) and high-performance mathematical functions. It is a core component of the Python data science ecosystem, serving as the backbone for other major libraries like Pandas, Matplotlib, and scikit-learn.

Matplotlib :

Matplotlib is a foundational, powerful Python library for creating a wide range of static, animated, and interactive visualizations in data science. It provides extensive customization options and serves as the backbone for many other visualization libraries like Seaborn.

Seaborn :

Seaborn is a powerful Python library for statistical data visualization, built on Matplotlib, that creates attractive, informative plots with minimal code, seamlessly integrating with pandas DataFrames to reveal trends, relationships, and patterns in data, making it essential for Exploratory Data Analysis (EDA). It offers high-level functions for complex plots (like distributions, regressions, heatmaps, violin plots) with beautiful default styles, color palettes, and themes, simplifying the process of understanding data structures and distributions.

Scikit-learn :

Scikit-learn (sklearn) is the most popular, open-source Python library for classical machine learning, providing simple and efficient tools for predictive data analysis. It is built on top of the scientific Python stack (NumPy, SciPy, Matplotlib) and is an essential tool for data scientists.

Ch-4: Understanding data :

4.1: Importing necessary Python libraries :

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

4.2: Creating the data frame :

```
df = pd.read_excel("D:\\VNSGU_PDA_DATA\\Data-Analytics-using-Python-Minor-Project\\Data\\raw\\VNSGU_Results.xlsx")
```

4.3 display all column names :

```
print(df.columns)
```

✓ 0.0s

```
Index(['Seat Number', 'Name', 'College Name', 'Total Marks', 'SGPA', 'Result',
      'Course_Name', 'ADVANCE_WEB_TECHNOLOGY_Ext_Theory',
      'ADVANCE_WEB_TECHNOLOGY_Ext_Practical',
      'ADVANCE_WEB_TECHNOLOGY_Total_Ext', 'ADVANCE_WEB_TECHNOLOGY_Int_Theory',
      'ADVANCE_WEB_TECHNOLOGY_Int_Practical',
      'ADVANCE_WEB_TECHNOLOGY_Total_Int', 'ADVANCE_WEB_TECHNOLOGY_Total_Obt',
      'WEB_FRAMEWORK_AND_SERVICES_Ext_Theory',
      'WEB_FRAMEWORK_AND_SERVICES_Ext_Practical',
      'WEB_FRAMEWORK_AND_SERVICES_Total_Ext',
      'WEB_FRAMEWORK_AND_SERVICES_Int_Theory',
      'WEB_FRAMEWORK_AND_SERVICES_Int_Practical',
      'WEB_FRAMEWORK_AND_SERVICES_Total_Int',
      'WEB_FRAMEWORK_AND_SERVICES_Total_Obt', '.NET_TECHNOLOGY_Ext_Theory',
      '.NET_TECHNOLOGY_Ext_Practical', '.NET_TECHNOLOGY_Total_Ext',
      '.NET_TECHNOLOGY_Int_Theory', '.NET_TECHNOLOGY_Int_Practical',
      '.NET_TECHNOLOGY_Total_Int', '.NET_TECHNOLOGY_Total_Obt',
      'LINUX_OPERATING_SYSTEM(LOS)_Ext_Theory',
      'LINUX_OPERATING_SYSTEM(LOS)_Ext_Practical',
      'LINUX_OPERATING_SYSTEM(LOS)_Total_Ext',
      'LINUX_OPERATING_SYSTEM(LOS)_Int_Theory',
      'LINUX_OPERATING_SYSTEM(LOS)_Int_Practical',
      'LINUX_OPERATING_SYSTEM(LOS)_Total_Int',
      'LINUX_OPERATING_SYSTEM(LOS)_Total_Obt', 'NETWORK_TECHNOLOGY_Total_Ext',
      'NETWORK_TECHNOLOGY_Total_Int', 'NETWORK_TECHNOLOGY_Total_Obt',
      'Course_Total_Ext', 'Course_Total_Int', 'Course_Total_Obt'],
      dtype='object')
```

4.5 display top few lines of data :

```
df.head(5)
```

```
      Seat Number      Name \
0      1001  ADEP SAIKRISHNA KISHORBHAI
1      1002      AHIR KRISH PRAMODKUMAR
2      1003      AHIR MEET MANSUKHBHAI
3      1004      AHIR MITKUMAR SURESHBHAI
4      1005      AHIR NILAY ARVINDBHAI

      College Name Total Marks  SGPA Result \
0  V.S. PATEL COLLEGE OF ARTS & SCIENCE, BILIMORA  332 / 550  6.18  PASS
1  V.S. PATEL COLLEGE OF ARTS & SCIENCE, BILIMORA  347 / 550  6.36  PASS
2  V.S. PATEL COLLEGE OF ARTS & SCIENCE, BILIMORA  348 / 550  6.18  PASS
3  V.S. PATEL COLLEGE OF ARTS & SCIENCE, BILIMORA  347 / 550  6.36  PASS
4  V.S. PATEL COLLEGE OF ARTS & SCIENCE, BILIMORA  325 / 550  5.82  PASS

      Course_Name \
0  CERTIFICATE COURSE ON GOOGLE PRODUCTS(CCGP)
1  CERTIFICATE COURSE ON GOOGLE PRODUCTS(CCGP)
2  CERTIFICATE COURSE ON GOOGLE PRODUCTS(CCGP)
3  CERTIFICATE COURSE ON GOOGLE PRODUCTS(CCGP)
4  CERTIFICATE COURSE ON GOOGLE PRODUCTS(CCGP)
```

```
ADVANCE_WEB_TECHNOLOGY_Ext_Theory  ADVANCE_WEB_TECHNOLOGY_Ext_Practical \
0                                11.0                                22.0
1                                 9.0                                19.0
2                                16.0                                19.0
...
3                                25.0                                14          39.0
4                                20.0                                11          31.0

[5 rows x 41 columns]
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

4.6 display shape of data :

```
df.shape
```

```
(8987, 49)
```

4.7 Getting summary of the dataframe using df.info() :

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8987 entries, 0 to 8986
Data columns (total 41 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Seat Number                          8987 non-null   int64
1   Name                                8987 non-null   object
2   College Name                        8987 non-null   object
3   Total Marks                        8987 non-null   object
4   SGPA                               8987 non-null   object
5   Result                             8987 non-null   object
6   Course_Name                        8987 non-null   object
7   ADVANCE_WEB_TECHNOLOGY_Ext_Theory  8950 non-null   float64
8   ADVANCE_WEB_TECHNOLOGY_Ext_Practical 8805 non-null   float64
9   ADVANCE_WEB_TECHNOLOGY_Total_Ext    8793 non-null   float64
10  ADVANCE_WEB_TECHNOLOGY_Int_Theory    8987 non-null   int64
11  ADVANCE_WEB_TECHNOLOGY_Int_Practical 8942 non-null   float64
12  ADVANCE_WEB_TECHNOLOGY_Total_Int     8942 non-null   float64
13  ADVANCE_WEB_TECHNOLOGY_Total_Obt     8987 non-null   int64
14  WEB_FRAMEWORK_AND_SERVICES_Ext_Theory 8948 non-null   float64
15  WEB_FRAMEWORK_AND_SERVICES_Ext_Practical 8830 non-null   float64
16  WEB_FRAMEWORK_AND_SERVICES_Total_Ext  8819 non-null   float64
17  WEB_FRAMEWORK_AND_SERVICES_Int_Theory 8986 non-null   float64
18  WEB_FRAMEWORK_AND_SERVICES_Int_Practical 8940 non-null   float64
19  WEB_FRAMEWORK_AND_SERVICES_Total_Int  8940 non-null   float64
...
40  Course_Total_Obt                     8986 non-null   float64
dtypes: float64(31), int64(4), object(6)
```

Ch-5: Understanding spread of data :

With our dataset we have used histogram with “kde=True” to understand the distribution and spread of data in column “rate” (user ratings).

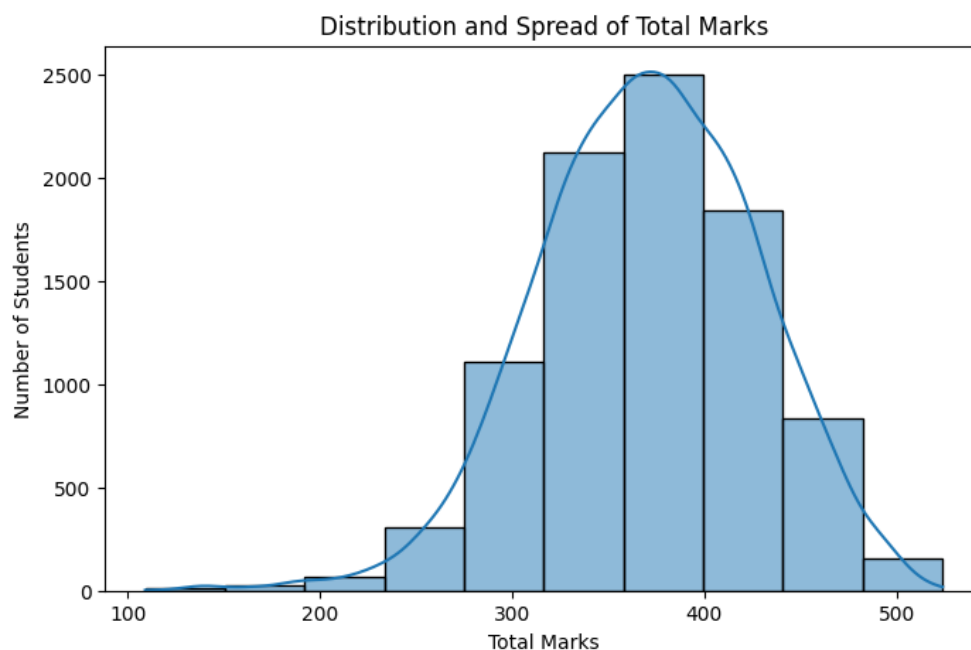
5.1 Distribution of Total Marks (Histogram)

To understand how students performed overall, we visualize the distribution of total marks using a histogram with kernel density estimation (KDE).

A histogram shows:

- How marks are distributed
- Which score range has maximum students
- Whether distribution is normal, skewed, or uneven

```
</> Python
plt.figure(figsize=(8,5))
sns.histplot(df['Total'], bins=10, kde=True)
plt.title("Distribution and Spread of Total Marks")
plt.xlabel("Total Marks")
plt.ylabel("Number of Students")
plt.show()
```



5.2 Spread of Subject-wise Marks (Boxplot Analysis)

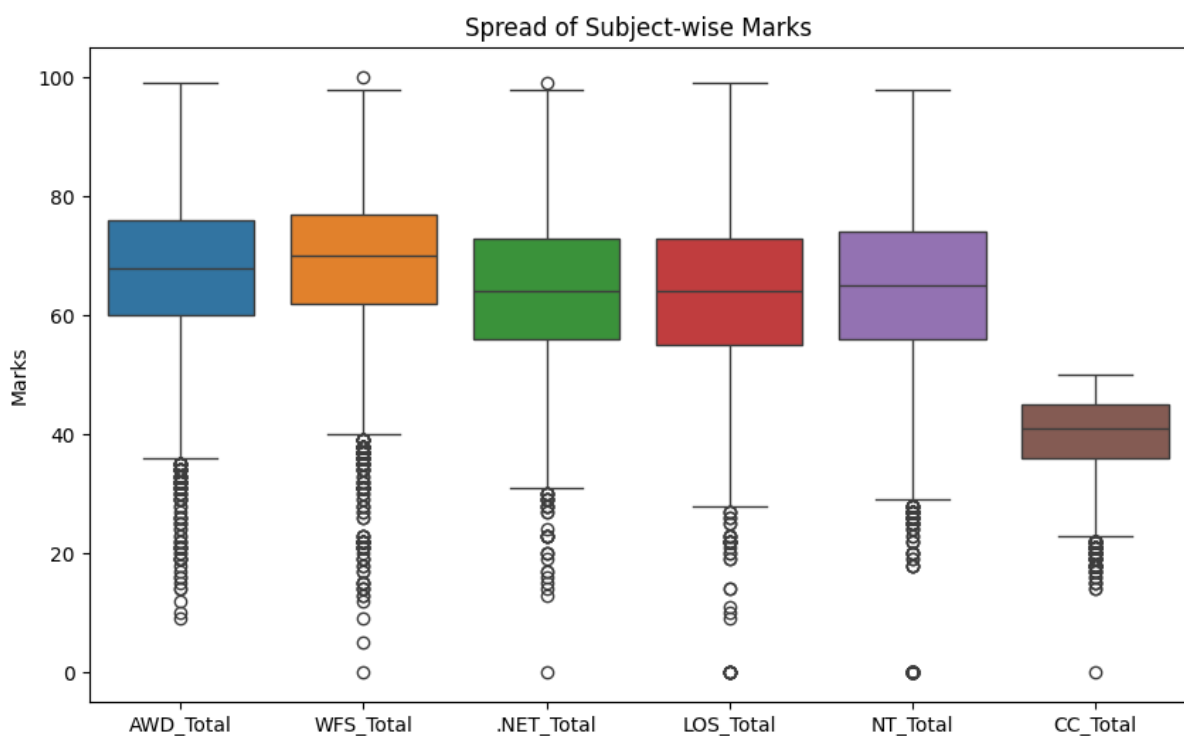
To understand variability and detect outliers in individual subjects, we use boxplots.

A boxplot helps in:

- Identifying median marks
- Observing spread (IQR)
- Detecting outliers
- Comparing subject difficulty levels

`</> Python`

```
plt.figure(figsize=(10,6))
sns.boxplot(data=df[['AWD_Total','WFS_Total','NET_Total','LOS_Total','NT_Total','CC_Total']])
plt.title("Spread of Subject-wise Marks")
plt.ylabel("Marks")
plt.show()
```



Ch-6: Automating EDA using python :

Automating Exploratory Data Analysis (EDA) in Python is achieved using specialized libraries that generate comprehensive, interactive reports with just a few lines of code.

In Python's pandas library, `df.info()` and `df.describe()` are fundamental methods for performing initial exploratory data analysis (EDA).

df.info(): Provides a concise structural summary of the DataFrame. It provides following information about dataset:

- Data Types (Dtypes),
- Non-Null Counts,
- Column Names,
- Memory Usage,
- Total Entries/Rows

df.describe(): Offers a statistical summary of the numerical data.

- Count: The number of non-null observations in each column.
- Mean: The average value.
- Std: The standard deviation (spread of data).
- Min: The minimum value.
- 25%, 50%, 75%: The first quartile, median (50th percentile), and third quartile.
- Max: The maximum value.

These functions help data analysts quickly understand the dataset's composition and spot potential issues like missing values or outliers without manually sifting through rows.

6.1: Getting summary of the dataframe using df.info()

Python

```
print("----- Dataset Structural Summary -----")
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8987 entries, 0 to 8986
Data columns (total 49 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Seat Number                               8987 non-null   int64
1   Name                                       8987 non-null   object
2   College Name                             8987 non-null   object
3   Total Marks                              8987 non-null   object
4   SGPA                                      8987 non-null   object
5   Result                                    8987 non-null   object
6   Course_Name                              8987 non-null   object
7   ADVANCE_WEB_TECHNOLOGY_Ext_Theory         8950 non-null   float64
8   ADVANCE_WEB_TECHNOLOGY_Ext_Practical       8805 non-null   float64
9   ADVANCE_WEB_TECHNOLOGY_Total_Ext          8793 non-null   float64
10  ADVANCE_WEB_TECHNOLOGY_Int_Theory          8987 non-null   int64
11  ADVANCE_WEB_TECHNOLOGY_Int_Practical       8942 non-null   float64
12  ADVANCE_WEB_TECHNOLOGY_Total_Int           8942 non-null   float64
13  ADVANCE_WEB_TECHNOLOGY_Total_Obt           8987 non-null   int64
14  WEB_FRAMEWORK_AND_SERVICES_Ext_Theory      8948 non-null   float64
15  WEB_FRAMEWORK_AND_SERVICES_Ext_Practical    8830 non-null   float64
16  WEB_FRAMEWORK_AND_SERVICES_Total_Ext       8819 non-null   float64
17  WEB_FRAMEWORK_AND_SERVICES_Int_Theory      8986 non-null   float64
18  WEB_FRAMEWORK_AND_SERVICES_Int_Practical    8940 non-null   float64
19  WEB_FRAMEWORK_AND_SERVICES_Total_Int       8940 non-null   float64
...
47  Course_Total_Int                           8987 non-null   int64
48  Course_Total_Obt                           8986 non-null   float64
```

6.2: Getting description of the dataframe using df.describe()

<> Python



```
print("----- Statistical Summary -----")
print(df.describe())
```

	Seat Number	Total Marks	SGPA	AWD_Ext_Th	AWD_Ext_Pr	\
count	8987.000000	8987.000000	8987.000000	8987.000000	8987.000000	
mean	5520.181262	370.020363	6.468580	12.560588	17.758874	
std	2609.081077	57.192697	1.689834	4.646001	4.240496	
min	1001.000000	109.000000	0.000000	0.000000	0.000000	
25%	3264.500000	332.000000	6.000000	9.000000	16.000000	
50%	5513.000000	371.000000	6.730000	13.000000	18.000000	
75%	7771.500000	410.000000	7.450000	16.000000	20.000000	
max	10042.000000	524.000000	9.820000	25.000000	25.000000	
	AWD_Ext_Total	AWD_Int_Th	AWD_Int_Pr	AWD_Int_Total	AWD_Total	\
count	8987.000000	8987.000000	8987.000000	8987.000000	8987.000000	
mean	30.148103	17.628463	19.849227	37.415266	67.823968	
std	7.848625	3.600709	3.735655	6.508946	11.734305	
min	0.000000	5.000000	0.000000	0.000000	9.000000	
25%	26.000000	15.000000	18.000000	34.000000	60.000000	
50%	31.000000	18.000000	20.000000	38.000000	68.000000	
75%	35.000000	20.000000	23.000000	42.000000	76.000000	
max	49.000000	25.000000	25.000000	50.000000	99.000000	
	...	LOS_Int_Pr	LOS_Int_Total	LOS_Total	NT_Ext_Total	\
count	...	8987.000000	8987.000000	8987.000000	8987.000000	
mean	...	17.958496	35.452988	63.905530	28.364527	
std	...	4.100120	6.918373	12.854728	9.659826	
min	...	0.000000	0.000000	0.000000	0.000000	
...						
75%		74.545455				
max		95.272727				

Ch-7: Handling missing data and outliers (Data cleaning and preparation) :

Before performing deeper analysis, it is essential to clean and prepare the dataset. Raw academic data may contain:

- Missing values
- Incorrect data types
- Duplicate records
- Extreme values (outliers)

Data cleaning ensures accurate and reliable analysis results.

7.1 Handling Missing Values :

Missing values can distort statistical calculations and visualizations. Therefore, we first identify and handle them properly.

 Python

```
print("----- Checking Missing Values -----")  
print(df.isnull().sum())
```

Seat Number	0
Name	0
College Name	0
Total Marks	0
SGPA	0
Result	0
Course_Name	0
ADVANCE_WEB_TECHNOLOGY_Ext_Theory	37
ADVANCE_WEB_TECHNOLOGY_Ext_Practical	182
ADVANCE_WEB_TECHNOLOGY_Total_Ext	194
ADVANCE_WEB_TECHNOLOGY_Int_Theory	0
ADVANCE_WEB_TECHNOLOGY_Int_Practical	45
ADVANCE_WEB_TECHNOLOGY_Total_Int	45
ADVANCE WEB TECHNOLOGY Total Obt	0

```

WEB_FRAMEWORK_AND_SERVICES_Ext_Theory      39
WEB_FRAMEWORK_AND_SERVICES_Ext_Practical    157
WEB_FRAMEWORK_AND_SERVICES_Total_Ext        168
WEB_FRAMEWORK_AND_SERVICES_Int_Theory        1
WEB_FRAMEWORK_AND_SERVICES_Int_Practical     47
WEB_FRAMEWORK_AND_SERVICES_Total_Int         47
WEB_FRAMEWORK_AND_SERVICES_Total_Obt         1
.NET_TECHNOLOGY_Ext_Theory                   39
.NET_TECHNOLOGY_Ext_Practical                103
.NET_TECHNOLOGY_Total_Ext                   113
.NET_TECHNOLOGY_Int_Theory                   1
...
NETWORK_TECHNOLOGY_Total_Obt                 49
Course_Total_Ext                             31
Course_Total_Int                             0
Course_Total_Obt                             1
dtype: int64

```

7.2: Fill Missing Values :

7.2.1 Handling Missing Marks :

During data inspection, certain subject columns contained missing values. Upon verification with examination records, these missing values represented students who scored zero marks in specific components (e.g., absent).

<> Python

```

# Convert selected columns to numeric first
df[columns_to_fill] = df[columns_to_fill].apply(pd.to_numeric, errors='coerce')

# Then fill NaN with 0
df[columns_to_fill] = df[columns_to_fill].fillna(0)

```

7.2.2 Handling SGPA Column :

The SGPA column contained the value "--" for students who failed the examination. Since this does not represent zero SGPA but rather an undefined grade point, these values were converted to NaN instead of zero to maintain statistical integrity.

 Python



```
df['SGPA'] = df['SGPA'].replace('--', np.nan)
df['SGPA'] = pd.to_numeric(df['SGPA'], errors='coerce')
```

7.2.3 Cleaning Total Marks Column :

The "Total Marks" column contained values in the format "Obtained Marks / Maximum Marks" (e.g., 345/550). Since only the obtained marks were required for analysis, the denominator part was removed.

 Python



```
df['Total Marks'] = df['Total Marks'].astype(str).str.split('/').str[0].str.strip()
df['Total Marks'] = pd.to_numeric(df['Total Marks'], errors='coerce')
```

7.2.4 Converting Marks Columns to Numeric :

The subject mark columns contained some non-numeric values and blank entries. Since these represented either absent students or zero marks, they were converted into numeric format and missing values were replaced with 0.

 Python



```
df[marks_columns] = (
    df[marks_columns]
    .apply(pd.to_numeric, errors='coerce')
    .fillna(0)
    .astype(int)
)
```

Ch-8 Univariate Analysis :

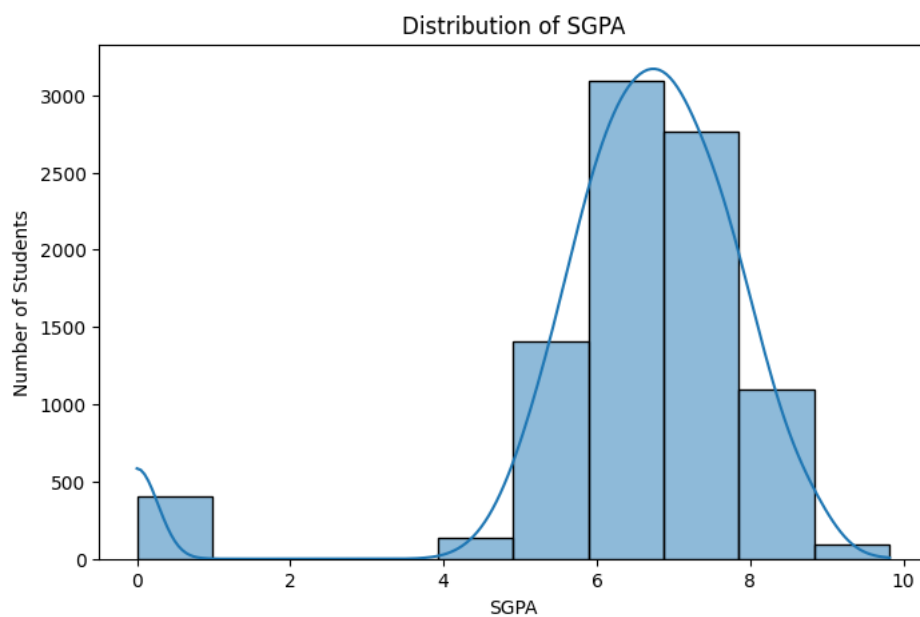
Univariate Analysis involves analyzing a single variable at a time to understand its distribution, central tendency, and frequency patterns.

In this chapter, we analyze:

- SGPA Distribution
- Pass vs Fail Ratio
- Subject-wise Failure Count
- Percentage Distribution

8.1 SGPA Distribution :

```
Python  
  
plt.figure(figsize=(8,5))  
sns.histplot(df['SGPA'], bins=10, kde=True)  
plt.title("Distribution of SGPA")  
plt.xlabel("SGPA")  
plt.ylabel("Number of Students")  
plt.show()
```



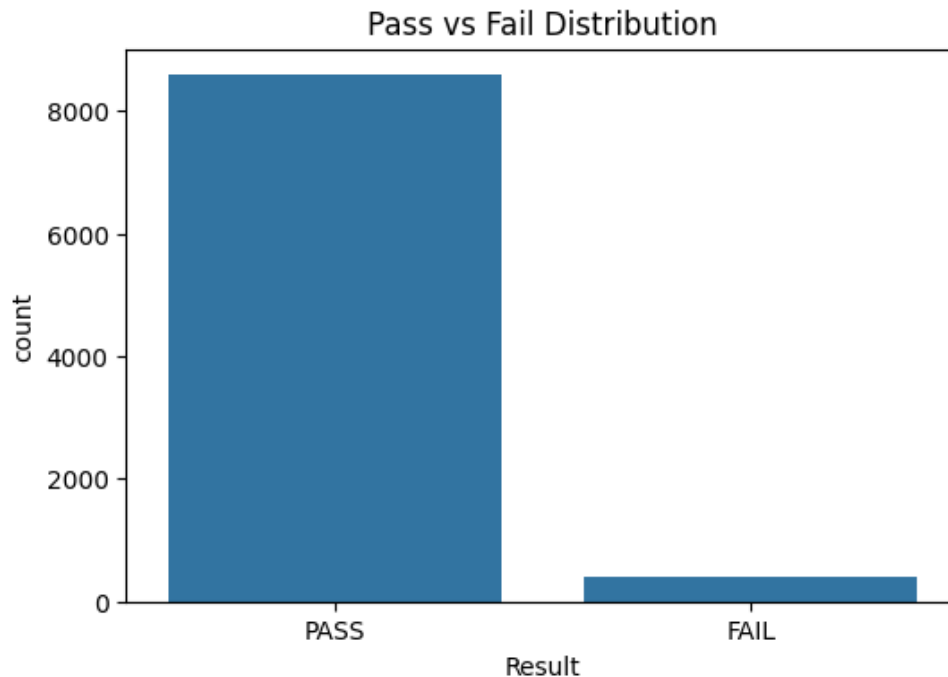
Interpretation:

- Majority clustered around 6–8 average performance.
- High peak near 8–9 strong academic performance.
- Values near 0 indicate failed students (if included).

8.2 Pass vs Fail Distribution :

Python

```
plt.figure(figsize=(6,4))
sns.countplot(x='Result', data=df)
plt.title("Pass vs Fail Distribution")
plt.show()
```



Interpretation:

- Higher PASS count → strong semester performance.
- Low FAIL/ATKT count → difficult subjects.

8.4 Failure Count Per Subject :

Python

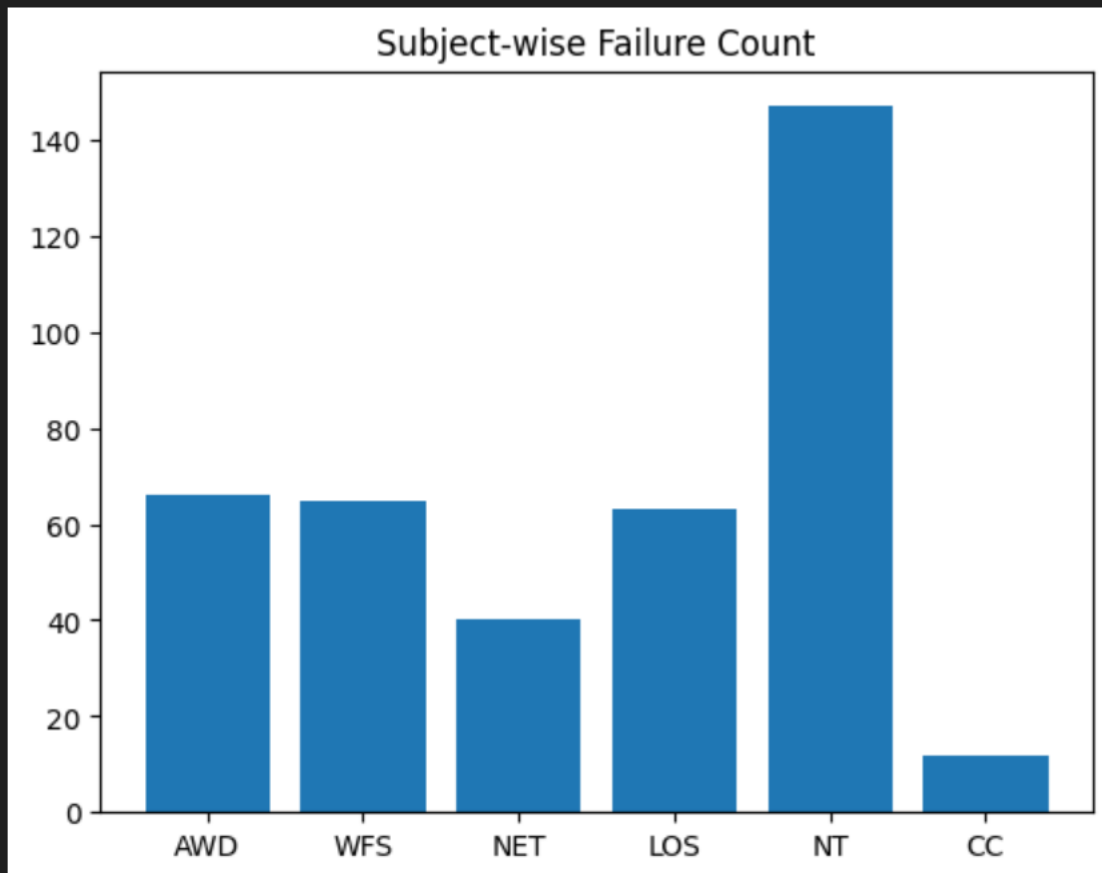
```
import matplotlib.pyplot as plt

# Subject-wise Failure Count based on Passing Criteria
failure_counts = {
    'AWD': (df['AWD_Total'] < 33).sum(),
    'WFS': (df['WFS_Total'] < 33).sum(),
    'NET': (df['.NET_Total'] < 33).sum(),
    'LOS': (df['LOS_Total'] < 33).sum(),
    'NT': (df['NT_Total'] < 33).sum(),
    'CC': (df['CC_Total'] < 18).sum(),
}

# Print Failure Counts
print("Subject-wise Failure Count:")
for subject, count in failure_counts.items():
    print(f"{subject}: {count}")

# Plot Bar Graph
plt.figure(figsize=(8,5))
plt.bar(failure_counts.keys(), failure_counts.values())
plt.title("Subject-wise Failure Count")
plt.xlabel("Subjects")
plt.ylabel("Number of Failed Students")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
{'AWD': 66, 'WFS': 65, 'NET': 40, 'LOS': 63, 'NT': 147, 'CC': 12}
```

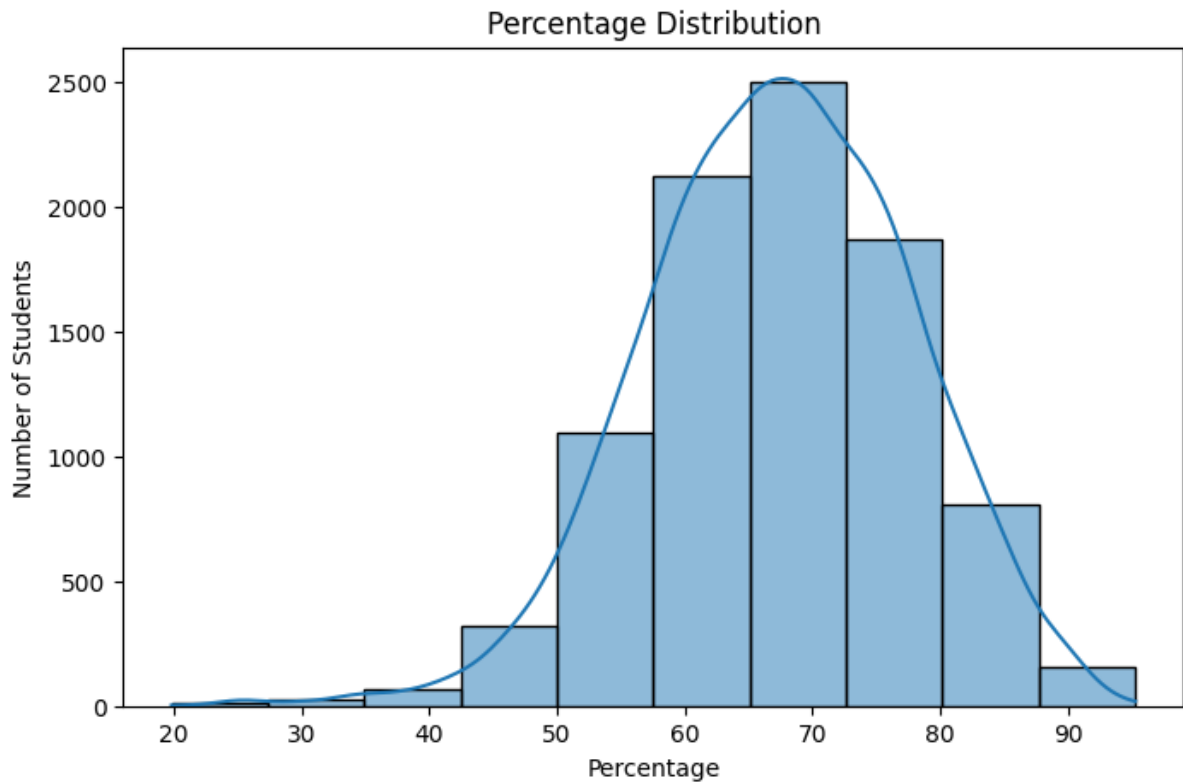


Interpretation:

- Network Technology (NT) has the highest failures (147), indicating it is the most difficult subject.
- AWD, WFS, and LOS show moderate failure counts, suggesting average difficulty.
- Cloud Computing (CC) has the lowest failures (12), showing strongest student performance.

8.6 Percentage Distribution :

```
</> Python
plt.figure(figsize=(8,5))
sns.histplot(df['Percentage'], bins=10, kde=True)
plt.title("Percentage Distribution")
plt.xlabel("Percentage")
plt.ylabel("Number of Students")
plt.show()
```



Interpretation :

- Most students scored between 55% and 75%, indicating average to good overall performance.
- The distribution appears approximately bell-shaped (normal), showing balanced academic results.
- Very few students scored below 40% or above 90%, indicating limited extreme performance cases.

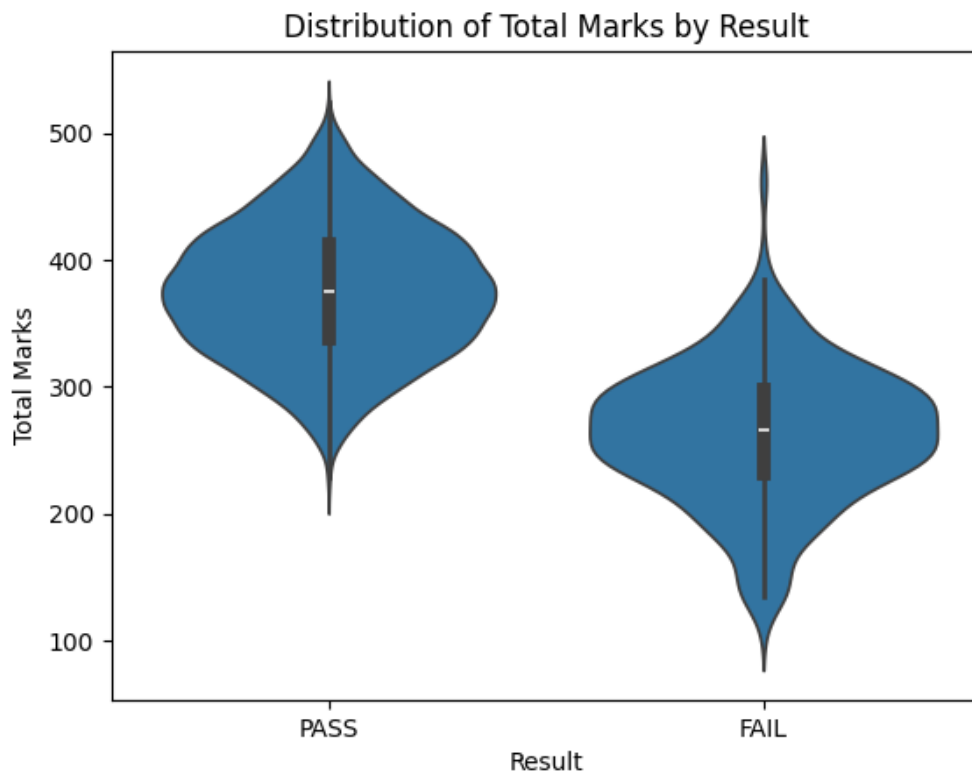
Ch-9: Bivariate Analysis

Bivariate Analysis involves examining the relationship between two variables to understand patterns, dependencies, and performance trends within the dataset. In this chapter, we analyze relationships between academic metrics such as total marks, percentage, result status, subject performance, and institutional comparison.

9.1 Distribution of Total Marks by Result :

Python

```
sns.violinplot(x='Result', y='Total Marks', data=df)
plt.title("Distribution of Total Marks by Result")
plt.show()
```



Interpretation:

The violin plot shows a clear distinction between PASS and NOT PASS students. Students who passed are concentrated in the higher total marks range, while non-passing students are clustered in the lower score region. The separation confirms that total marks strongly influence final result status.

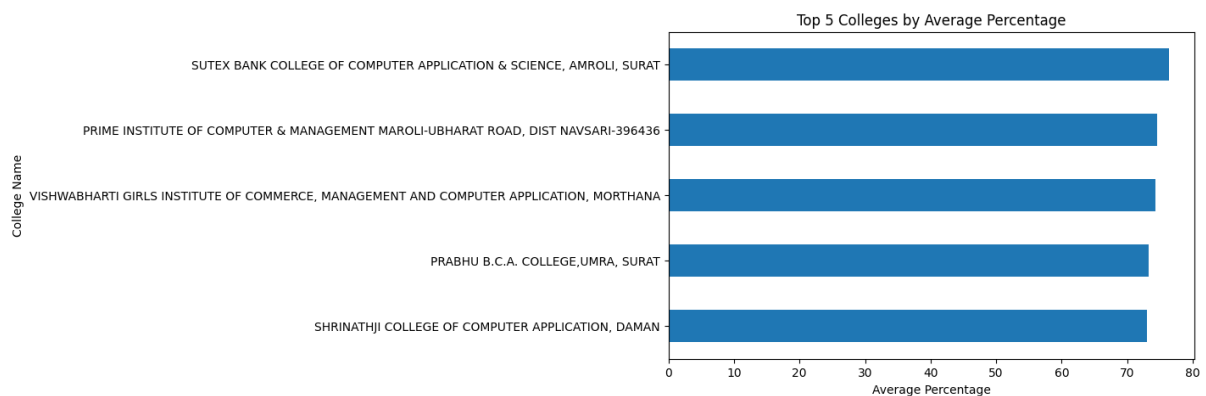
9.2 Top 5 Colleges by Average Percentage :

`</>` Python

```
college_avg = df.groupby('College Name')['Percentage'].mean().sort_values(ascending=False)

top5 = college_avg.head(5)

plt.figure(figsize=(8,5))
top5.sort_values().plot(kind='barh')
plt.title("Top 5 Colleges by Average Percentage")
plt.xlabel("Average Percentage")
plt.tight_layout()
plt.show()
```



Interpretation:

The comparison reveals measurable differences in academic performance across institutions. The top-performing colleges demonstrate consistently higher average percentages, indicating stronger academic outcomes. This analysis highlights institutional variation under the same university framework.

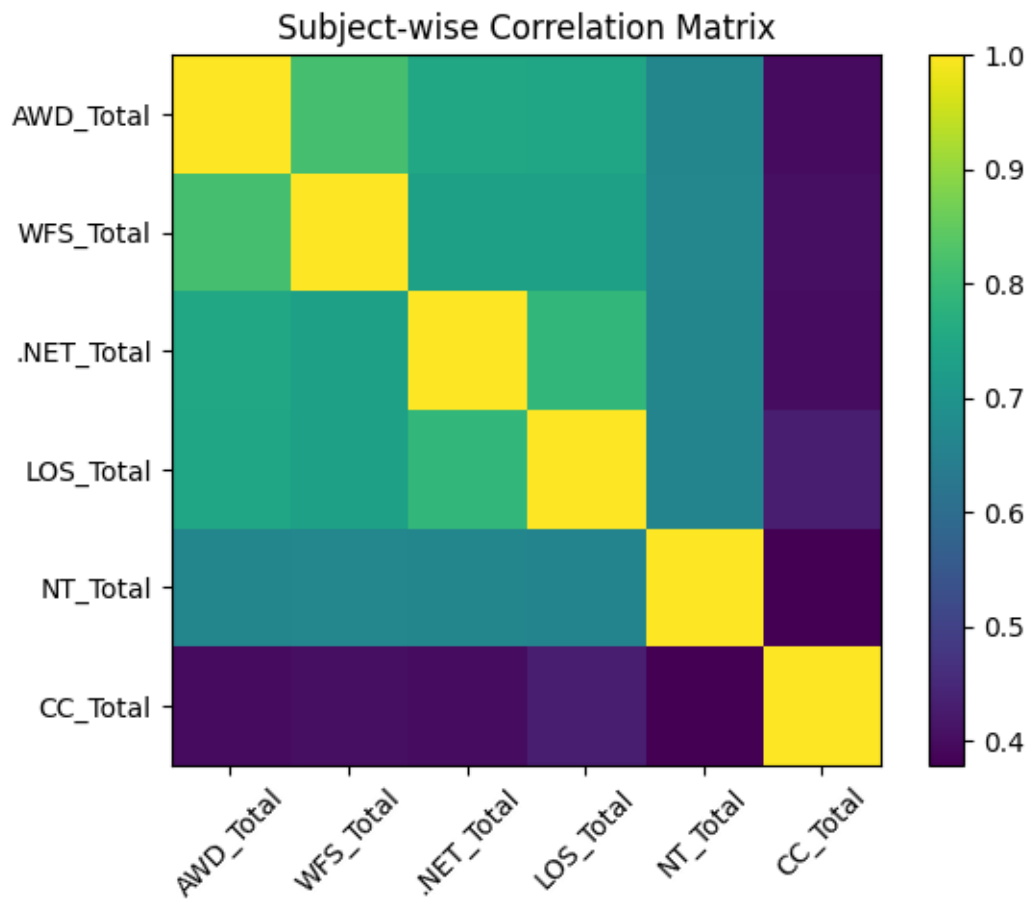
9.3 Subject-wise Correlation Analysis :

`</>` Python

```
subject_cols = ['AWD_Total', 'WFS_Total', '.NET_Total', 'LOS_Total', 'NT_Total', 'CC_Total']

corr_matrix = df[subject_cols].corr()

plt.figure()
plt.imshow(corr_matrix)
plt.xticks(range(len(subject_cols)), subject_cols, rotation=45)
plt.yticks(range(len(subject_cols)), subject_cols)
plt.title("Subject-wise Correlation Matrix")
plt.colorbar()
plt.show()
```



Interpretation:

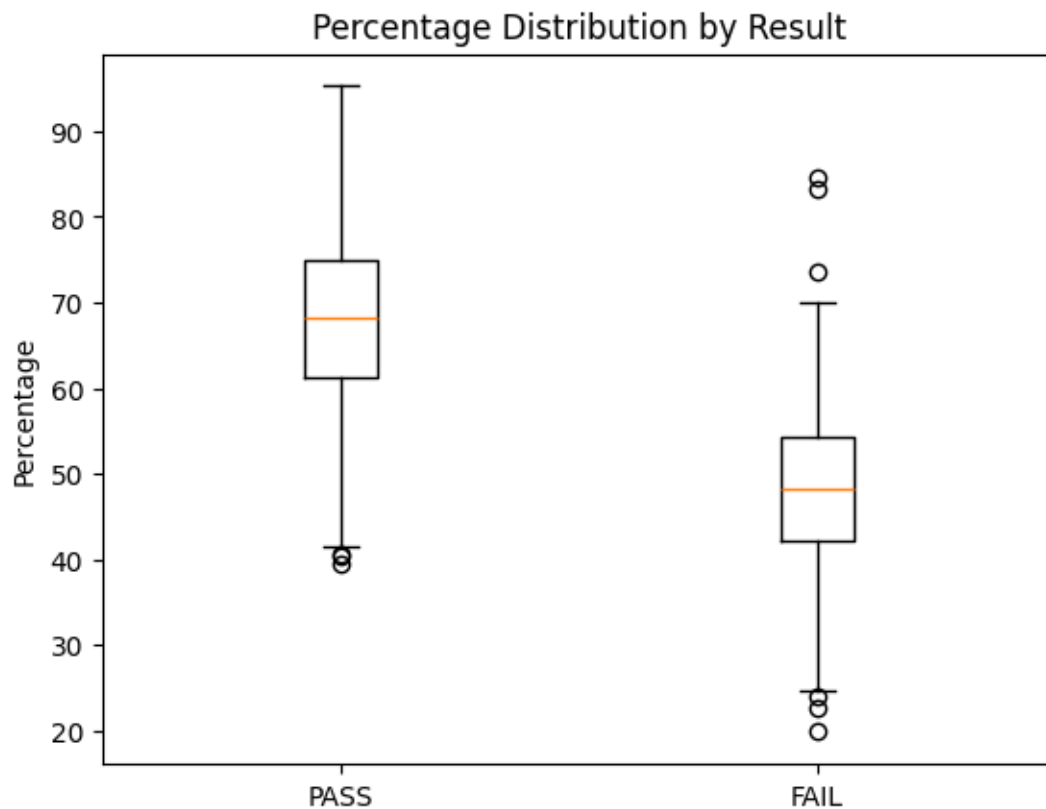
The correlation heatmap indicates strong positive relationships among several theory-based subjects. High correlation values suggest that students who perform well in one subject tend to perform well in related subjects. Lower correlation values indicate relatively independent performance patterns.

9.4 Percentage Distribution by Result :

Python

```
pass_percent = df[df['Result'] == 'PASS']['Percentage']
fail_percent = df[df['Result'] != 'PASS']['Percentage']

plt.boxplot([pass_percent, fail_percent], labels=['PASS', 'NOT PASS'])
plt.title("Percentage Distribution by Result")
plt.ylabel("Percentage")
plt.show()
```



Interpretation:

The median percentage of PASS students is significantly higher than that of NOT PASS students. The boxplot clearly shows that percentage is a strong determinant of final academic outcome. The variability among failing students is greater, indicating inconsistent performance patterns.

9.5 Subject-wise Passing Percentage :

Python

```
# Total number of students
total_students = len(df)

# Calculate passing percentage per subject
passing_percentage = {
    'AWD': (df['AWD_Total'] >= 33).sum() / total_students * 100,
    'WFS': (df['WFS_Total'] >= 33).sum() / total_students * 100,
    'NET': (df['.NET_Total'] >= 33).sum() / total_students * 100,
    'LOS': (df['LOS_Total'] >= 33).sum() / total_students * 100,
    'NT': (df['NT_Total'] >= 33).sum() / total_students * 100,
    'CC': (df['CC_Total'] >= 18).sum() / total_students * 100 # CC passing criteria
}

# Create Bar Chart
plt.figure()
plt.bar(passing_percentage.keys(), passing_percentage.values())

plt.title("Subject-wise Passing Percentage")
plt.xlabel("Subjects")
plt.ylabel("Passing Percentage")

plt.ylim(0, 100) # Since percentage scale is 0-100
plt.show()
```

